



Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Assignment

- (1) Consider the equation $(1 + x^2)y'' - 4xy' + 6y = 0$.
- (a) Find its general solution in the form $y = a_0y_2(x) + a_1y_1(x)$, where $y_1(x)$ and $y_2(x)$ are power series solutions in powers of x .
- (b) Find the radius of convergence for $y_1(x)$ and $y_2(x)$. [4+1]
- (2) Find the first three **nonzero** terms in a power series expansion about $x_0 = \pi$ of the solution to the initial value problem

$$y'' - (\sin x)y = 0, \quad \text{with } y(\pi) = 1, \ y'(\pi) = 0.$$

[5]